Interval scheduling
1 second, 32 MB

In this task, we consider the interval scheduling problem where you have a single resource (think of a room) and (conflicting) requests to use that particular resource. You can accept a request but you have to ensure that you do not accept conflicting requests. You want to maximize the number of accepted requests.

You are given N intervals representing N requests. Each interval is a pair (s_i, t_i) such that $s_i < t_i$, where s_i is the request starting time and t_i is the request finishing time. Two intervals are *not compatible* if they *strictly* intersect. More specifically, intervals i and j are not compatible if $s_i < t_j$ and $s_j < t_i$. (Note that if they only share the end points, they are compatible.)

Input
The first line of the input contains an integer N ($1 <= N <= 100{,}000$). The next N lines describe the intervals. More specifically, line $1+i$, for $1 <= i <= N$, contains two integers s_i and t_i ($1 <= s_i < t_i <= 1{,}000{,}000{,}000$)

Output
The output contains a single integer, the maximum number of compatible intervals.

Examples

input:	output:
3 1 2 2 3 1 3	2
input: 4 1 2 1 4 3 4 2 4	output: 2